

An unknown population SD

When the population standard deviation σ is unknown, a mean interval estimates it with the sample standard deviation s and uses a t critical value.

Does a large sample make the population SD known?

An estimated standard error

An estimated standard error for a sample mean uses s in place of σ .

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}$$

For $s=5$ minutes and $n=25$, what is the estimated standard error?

A t statistic

A t-score (t statistic), like a z-score, measures standardized distance, but uses an estimated standard error.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

What extra estimated quantity distinguishes this from the known-SD z statistic?

The t distribution

Student's t-distribution is symmetric about zero and has heavier tails than the standard normal distribution.

What additional uncertainty might those heavier tails reflect?

Degrees of freedom

Degrees of freedom (df) count freely varying deviations. For a one-sample t procedure, $df = n - 1$.

$$df = n - 1$$

What degrees of freedom belong to a sample of 25 observations?

Why $n-1$?

For a sample mean, the deviations from that mean sum to zero.

If four of five deviations are known, can the fifth vary freely?

Changing degrees of freedom

As degrees of freedom increase, the t distribution approaches the standard normal.

Would $df=4$ or $df=100$ produce heavier tails?

A critical t value

A positive t^* value cuts off $\alpha/2$ in the upper tail of the appropriate t distribution.

Which degrees of freedom and left-tail area locate t^* for $n=25$ and 95% confidence?

Reading a t table

A t table may label columns by confidence, one-tail area, or two-tail area. Use $df = 14$ for a 95% interval.

What area should you locate in each kind of table?

Comparing critical values

Given: at 95% confidence and $df=24$, $t^*=2.064$. The standard-normal value is $z^*=1.96$.

How would the two values affect margins computed with the same estimated standard error?

Conditions for a t interval

Use independent random observations and a normal population, or a sample large enough for a reasonable approximation without problematic skewness or outliers.

Why would a small sample with an extreme outlier deserve caution?

A small normal sample

Synthetic independent random sample: $n=10$ from a normal population with unknown σ .

Is $n < 30$ by itself a reason to reject the t method?

A problematic small sample

Synthetic random sample: $n=8$, strong right skew, and one value far above all the others.

Would a t interval automatically be reliable?

Checking independence

A synthetic sample contains 25 consecutive measurements from one machine whose output drifts over time.

What assumption needs attention before using $s/\sqrt{n}$?

Choosing z or t

Synthetic independent random sample:
 $n=100$; the population SD is unknown and the sample SD is 5 minutes.

Which method estimates μ : known-SD z or one-sample t?

The t margin of error

The error bound for a population mean (EBM) uses a t critical value when σ is estimated by s .

$$EBM = t^* \frac{s}{\sqrt{n}}$$

For $t^*=2.064$, $s=5$, and $n=25$, what is EBM?

Constructing a t interval

Synthetic independent random sample: $n=25$, mean=12 minutes, $s=5$ minutes. Normal population; unknown σ ; 95% confidence; $t^*=2.064$.

What is the confidence interval for μ ?

Interpreting a t interval

Synthetic result: a 95% t interval for mean processing time is 9.94 to 14.06 minutes.

What population quantity does this interval estimate?

Individual spread and mean precision

Synthetic sample: $n=25$, sample $SD=5$ minutes, estimated standard error=1 minute.

Which number describes spread among observed processing times?

A higher confidence level

Use the same sample, s , and degrees of freedom. Compare 95% and 99% t intervals.

What changes in the interval formula?

A larger sample

Hold the observed mean, sample SD, and confidence level fixed while n increases.

How do standard error and the critical t value each respond?

A raw-data interval

Synthetic independent random sample from a normal population, in seconds: 18, 19, 20, 21, 22. Unknown σ ; 95% confidence; $t^*=2.776$.

What t interval for μ follows from these observations?

Retaining units

Synthetic sample mean and sample SD are both measured in seconds.

What units should the standard error, margin of error, and interval endpoints have?

The reported conclusion

Synthetic result: a 95% t interval for μ is 9.94 to 14.06 minutes.

Would this establish that the mean is exactly 12 minutes?

An interval and a claim

Synthetic result: a 95% t interval for μ is 9.94 to 14.06 minutes. A claim states $\mu=8$.

Is 8 minutes compatible with this interval?

Sketching the confidence interval

Synthetic $n = 16$ sample: a 95% t interval for μ is $(18, 22)$, with sample mean 20.

How would you mark the estimate, endpoints, and error bound, and show the critical tail areas on a t curve?